

Goal-Aware AutoML System for Tabular Data

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Final Project – Data Mining

1. Project Overview

This project presents a complete Goal-Aware AutoML framework designed to automate the entire machine learning pipeline for tabular datasets. Its primary innovation lies in interpreting the user's natural language goal, detecting the target column accordingly, and selecting appropriate task type, preprocessing methods, and supervised models automatically.

The system performs:

- Data cleaning (removing duplicates, constants, high-missing columns)
- Schema inference and column type detection
- Natural-language-informed target detection
- Automatic classification/regression task detection
- Multiple preprocessing strategy generation
- Exhaustive evaluation of model × preprocessing × hyperparameter pipelines
- Final model training and test evaluation
- Plot generation and summary reporting

The goal of the system is to simplify machine learning experimentation for users while ensuring transparent, reproducible, and high-quality results.

2. Problem Definition and Objectives

The project addresses a common real-world difficulty: users possess data and a goal but may not know how to identify the correct target variable or decide which machine learning techniques to apply. **Objectives:** 1. Load and clean the dataset automatically. 2. Detect the target variable using textual goal interpretation. 3. Determine whether the task is classification or regression. 4. Explore multiple preprocessing strategies with imputation, scaling, and encoding. 5. Train, cross-validate, and compare multiple models and hyperparameters. 6. Select and evaluate the best pipeline. 7. Generate visualizations and a clear textual report.

3. Theoretical Background

3.1 Data Preprocessing

The system applies generic cleaning before modeling: removing duplicates, replacing infinities, dropping constant and >95% missing columns, and trimming whitespace. During model pipelines, it tests three preprocessing configurations combining numeric imputation, scaling, and categorical encoding through OneHotEncoder.

3.2 Task Types

The AutoML system distinguishes between classification and regression using the distribution and type of the target variable.

3.3 Machine Learning Algorithms

The model portfolio includes: Logistic Regression, Decision Trees, Random Forests, KNN, Naive Bayes, SVM, and GPU-accelerated XGBoost for classification; and Linear Regression, Polynomial Regression, Decision Trees, Random Forests, KNN, and XGBoost for regression.

3.4 Evaluation Metrics

Classification metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC.

Regression metrics: MAE, MSE, RMSE, R^2 .

3.5 Cross-Validation

StratifiedKFold is used for classification and KFold for regression to compute consistent performance estimates.

4. Methodology

4.1 Tools and Libraries

Python, Scikit-learn, XGBoost, Pandas, NumPy, Seaborn, and Matplotlib.

4.2 Dataset Description

The dataset is loaded from CSV and cleaned using consistent, reproducible rules. The user can supply either a single file or separate train/test files.

4.3 Data Preprocessing and EDA

Column type inference, missingness analysis, id-like column detection, and schema summary are performed automatically. Then multiple preprocessing pipelines are prepared.

4.4 Model Exploration

For each model and preprocessing strategy, all hyperparameter combinations are evaluated via cross-validation.

4.5 Model Evaluation

The best pipeline is retrained on the full training set and evaluated on the test set. The system computes metrics and generates plots (confusion matrix, ROC/PR, regression diagnostics, and feature importance).

5. Results and Analysis

The AutoML system produces a ranked list of pipelines based on their cross-validation scores. It then trains the best-performing pipeline on the full training dataset and evaluates it on the test set. Visual outputs include:

- Confusion matrix or Predicted-vs-Actual chart
- ROC curve and Precision-Recall curve (if applicable)
- Feature importance horizontal bar plot

These results provide insights into both model performance and the explanatory power of features.

6. Discussion

The AutoML system successfully reduces the complexity of ML workflow design. Its main advantages include transparency, modularity, and ease of use. Limitations include runtime cost for large datasets and a limited feature-engineering scope. Future improvements may include PCA, Bayesian tuning, and automated report generation.

7. Conclusion

This project demonstrates a complete AutoML pipeline with natural-language target detection, schema inference, preprocessing strategy exploration, automated model selection, and full evaluation. It integrates key Data Mining concepts into a coherent, practical system that can adapt to many real-world tabular datasets.